

 Google: Gemma 4 31B (free)

[google/gemma-4-31b-it:free](#)



Model weights 

 Compare

 Playground

 Try this model

Gemma 4 31B Instruct is Google DeepMind's 30.7B dense multimodal model supporting text and image input with text output. Features a 256K token context window, configurable thinking/reasoning mode, native function calling, and ...

Show more 

MODALITIES

   → 

PRICE

Free

CONTEXT

262K

RELEASED

Apr 2, 2026

Providers

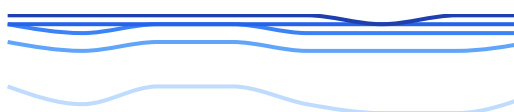
Different companies host the same model. OpenRouter routes your request to one of them based on the routing mode you pick — Balanced (price + speed), Nitro (fastest), or Exacto (highest tool-calling accuracy).

Free  		Latency / throughput				P50 
Provider 		 Input /M	 Output /M	 Latency	 Throughput	 Uptime
	Google AI Studio 	Free	Free	1.47s	25 tps	99.27%

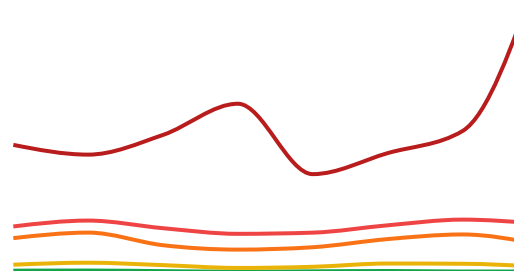
Performance

Throughput is how fast the model writes (tokens per second — higher is better). Latency is total round-trip time (lower is better). TTFT is time-to-first-token — how long before you see anything appear (lower is better).

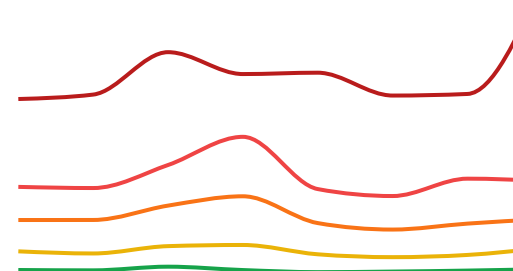
Throughput 25 tok/s P50, best across providers	Latency  1.47 s P50, best provider
All locations 	1 week 



P99	Avg 36 tok/s
P95	Avg 35 tok/s
P90	Avg 34 tok/s
P75	Avg 33 tok/s
P50	Avg 27 tok/s



P50	Avg 1.36 s
P75	Avg 1.81 s
P90	Avg 3.60 s
P95	Avg 4.73 s
P99	Avg 12.60 s

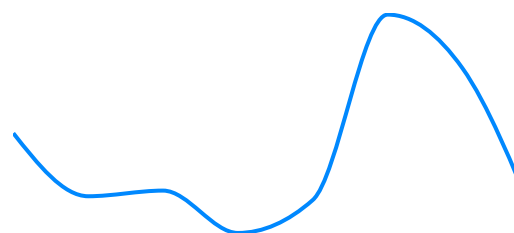


P50	Avg 7.22 s
P75	Avg 14.26 s
P90	Avg 27.06 s
P95	Avg 41.87 s
P99	Avg 82.06 s

AutoExacto Benchmarks

	GPQA Diamond	TAU-Bench
Together	84.2%	--
Together	84.0%	--
DeepInf...	83.2%	--
Samba...	82.5%	--

Tool Call Error Rate



Structured Output Error Rate

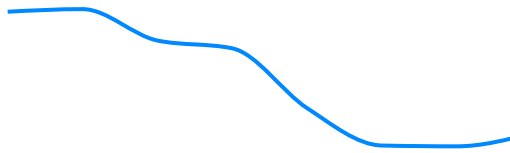
DeepInt... 84.1% 75.8%

ModelR... 83.3% 75.1%

DeepInf... 68.3% 75.9%

+12 more providers

Cache Hit Rate



Google AI Studio Avg 9.10 %



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Uptime is the percentage of the past 3 days that at least one provider was responding to requests.

Availability is the percentage of time that inference was successfully served. OpenRouter continuously monitors and uses the next-best provider when one returns an error.

Uptime (3d)

100.00%

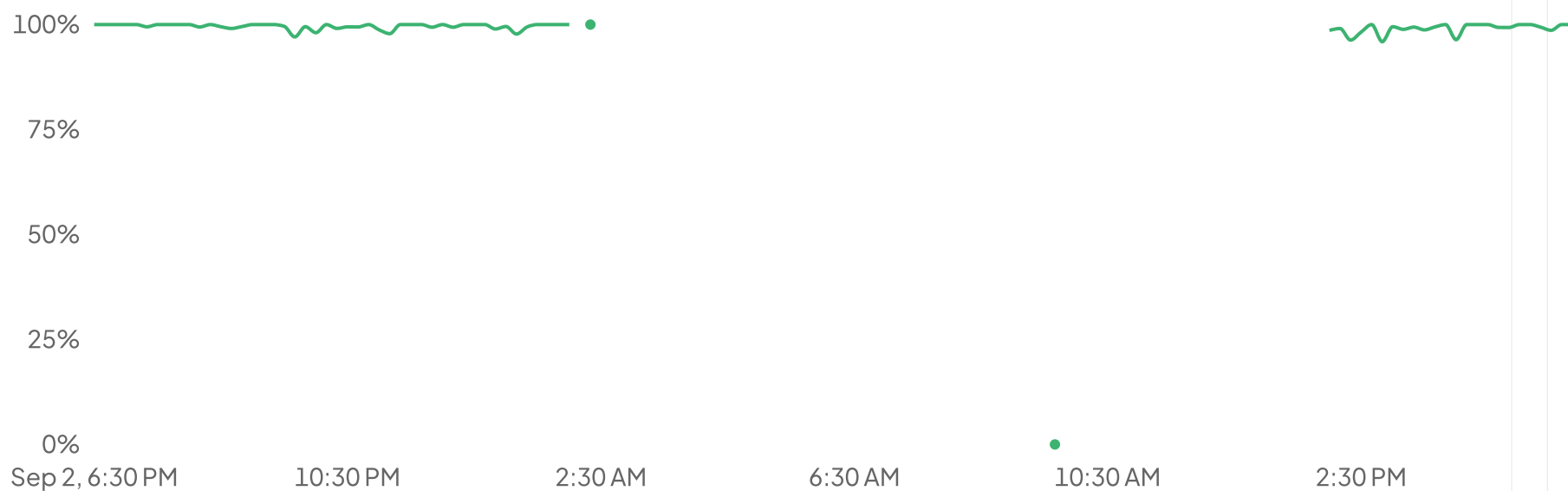
Availability (3d)

98.62%

Availability over the last 3 days Aug 31, 6 PM - Sep 3, 6 PM

Availability **98.62%**

Availability over the last 24 hours



When an error occurs in an upstream provider, we can recover by routing to another healthy provider, if your request filters allow it. You can access per-provider uptime data programmatically through the [Endpoints API](#). [Learn more](#) about our load balancing and customization options.

Benchmarks

Scores on standardized evaluations. Higher percentages are better — and rank percentile shows where this model lands among all models on OpenRouter.

Gemma 4 31B (Reasoning)

29.7

Artificial Analysis
Intelligence Index

Better than 34% of models
compared

43.4

Artificial Analysis
Coding Index

Better than 45% of models
compared

14.4

Artificial Analysis
Agentic Index

Better than 30% of models
compared

Reasoning

Coding

85.7%

HLE
Humanity's Last Exam

23.6%

IFBench
Instruction-following benchmark

75.6%

τ^2 -Bench Telecom
Conversational AI agents in dual-control scenarios

59.9%

AA-LCR
Long context reasoning evaluation

68.3%

GDPval-AA
Economically valuable tasks

15.7%

CritPt
Research-level physics reasoning

1.4%

45.4%

Terminal-Bench Hard
Agentic coding & terminal use

36.4%

Knowledge

Proportion of correctly answered questions

20.0%

AA-Omniscience Non-Hallucination Rate

Rate of avoiding hallucination among non-correct
responses

15.0%

Metrics sourced from [Artificial Analysis](#)

Apps

Public apps that send the most traffic to this model. Good signal for what real production workloads look like — and a hint at which use cases this model is best suited for.

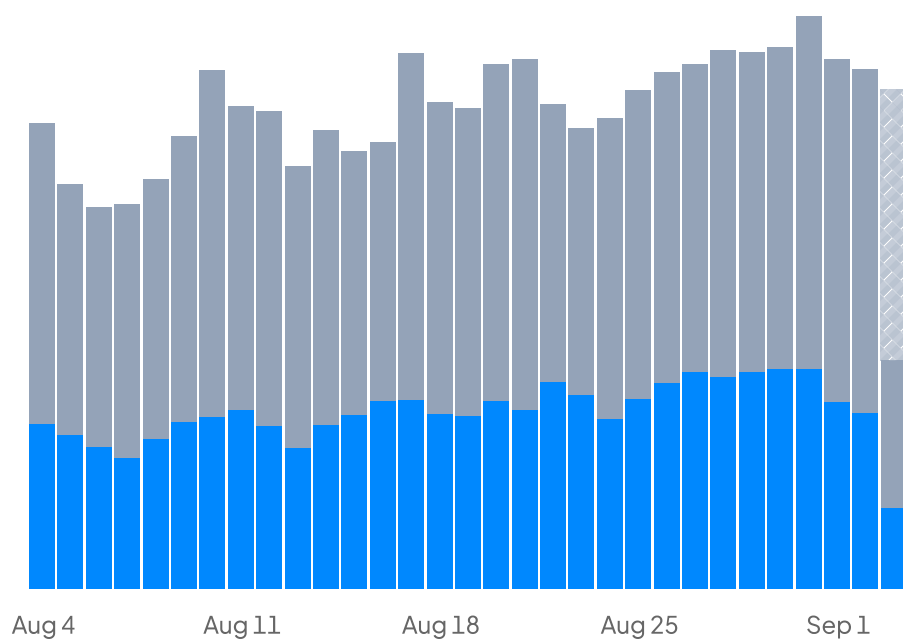
-  Janitor AI
Janitor AI is a chatbot platform where users create and chat with custo...
1.82B tokens
-  OpenClaw
OpenClaw is an open-source AI agent that connects to your messagin...
472M tokens
-  RiskForge Capital
new
218M tokens
-  LangChain

5.

**Chub AI**

GenAI for everyone

125M tokens



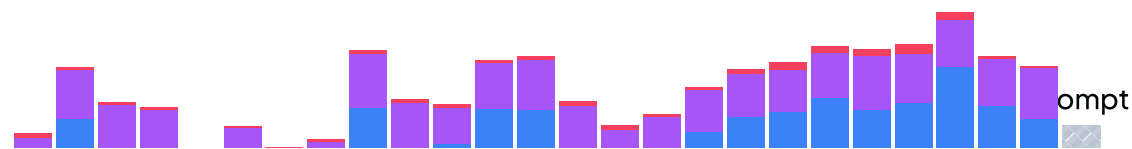
Activity

Token volume and request traffic to this model over time.

Tokens

300M

225M

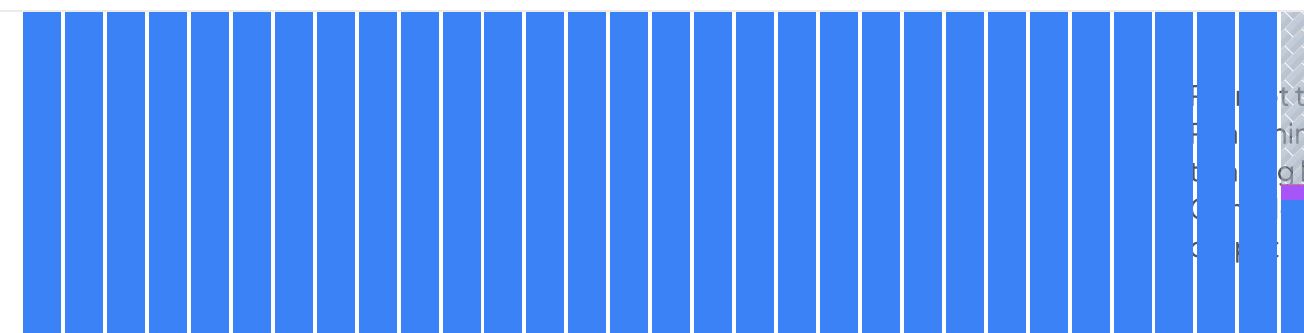


ompt

71.3M

150M

75M



Aug 4

Aug 8

Aug 12

Aug 16

Aug 20

Aug 24

Aug 28

Sep 1

Prompt tokens measure input size.
Thinking tokens show internal reasoning before a response.
Completion tokens reflect total output length.

Quick Start

Drop-in code to call this model. OpenRouter's API is OpenAI-compatible — most SDKs work by just swapping the base URL. The only thing that changes between models is the model slug below.

1 Get your API key

Create an API key from your OpenRouter dashboard and set it as an environment variable:

Create API Key

```
export OPENROUTER_API_KEY=sk-or-v1- ...
```

Copy

2 Make your first request

Use `google/gemma-4-31b-it:free` with the OpenRouter API:

OpenRouter supports reasoning-enabled models that can show their step-by-step thinking process. Use the `reasoning` parameter in your request to enable reasoning, and access the `reasoning_details` array in the response to see the model's

In the examples below, the OpenRouter-specific headers are optional. Setting them allows your app to appear on the OpenRouter leaderboards.

TypeScript SDK Python TypeScript (fetch) cURL Python (OpenAI) TypeScript (OpenAI)

```
import { OpenRouter } from "@openrouter/sdk";

const openrouter = new OpenRouter({
  apiKey: "<OPENROUTER_API_KEY>"
});

// Stream the response to get reasoning tokens in usage
const stream = await openrouter.chat.send({
  chatRequest: {
    model: "google/gemma-4-31b-it:free",
    messages: [
      {
        role: "user",
```

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Using third-party SDKs

For information about using third-party SDKs and frameworks with OpenRouter, please see our [frameworks documentation](#).

3 Enable streaming

Add `"stream": true` to your request body to receive responses as server-sent events:

```
curl -N https://openrouter.ai/api/v1/chat/completions \
  -H "Content-Type: application/json" \
  -H "Authorization: Bearer $OPENROUTER_API_KEY" \
  -d '{
    "model": "google/gemma-4-31b-it:free",
    "messages": [
      {
        role: "user",
```

Copy

```
{ "role": "user", "content": "Hello" }
]
```

Endpoint

Sends a request for a model response for the given chat conversation. Supports both streaming and non-streaming modes.

POST https://openrouter.ai/api/v1/chat/completions

Authorization	Bearer \$OPENROUTER_API_KEY
Content-Type	application/json
HTTP-Referer	optional – your site URL, for rankings
X-Title	optional – your site name, for rankings
Model	google/gemma-4-31b-it:free

Creates a streaming or non-streaming response using the OpenAI Responses API format. [Docs](#)

POST https://openrouter.ai/api/v1/responses

Authorization	Bearer \$OPENROUTER_API_KEY
Content-Type	application/json
HTTP-Referer	optional – your site URL, for rankings
X-Title	optional – your site name, for rankings
Model	google/gemma-4-31b-it:free

POST `https://openrouter.ai/api/v1/messages`

Authorization **Bearer \$OPENROUTER_API_KEY**

Content-Type **application/json**

HTTP-Referer optional – your site URL, for rankings

X-Title optional – your site name, for rankings

Model **google/gemma-4-31b-it:free**

Parameters

Name	Type	Default	Description
reasoning	map	—	Controls reasoning behavior for models that support thinking tokens, includ
max_tokens	integer	—	This sets the upper limit for the number of tokens the model can generate in r
temperature	float	1	This setting influences the variety in the model's responses.
top_p	float	0.95	This setting limits the model's choices to a percentage of likely tokens: only 1
seed	integer	—	If specified, the inferencing will sample deterministically, such that repeated
response_format	map	—	Forces the model to produce specific output format.
tools	array	—	Tool calling parameter, following OpenAI's tool calling request shape.
tool_choice	string or object	—	Controls which (if any) tool is called by the model.



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Frequently asked questions

What is Gemma 4 31B (free)?

Is Gemma 4 31B (free) free?

What is the context length of Gemma 4 31B (free)?

Does Gemma 4 31B (free) support tool calling and structured outputs?

What inputs and outputs does Gemma 4 31B (free) support?

What other text models does Google have?

When was Gemma 4 31B (free) released?

Explore more models

AI Models with Vision: Multimodal LLMs for Image Understanding Collection

AI Model Rankings Ranking

More models from Google

Gemini 3.8 Flash

Gemini 3.8 Flash is Google's most intelligent Flash model with significant ...

Text 1.0M context \$0.75 / 3.75

Gemini 3.8 Flash

Gemini 3.8 Flash is Google's most intelligent Flash model with significant ...

Text 1.0M context \$0.375 / 1.875

Gemini 3.7 Flash

Gemini 3.7 Flash is a multimodal moc from Google for fast agentic workflc

Text 1.0M context \$0.75 / 3.75



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